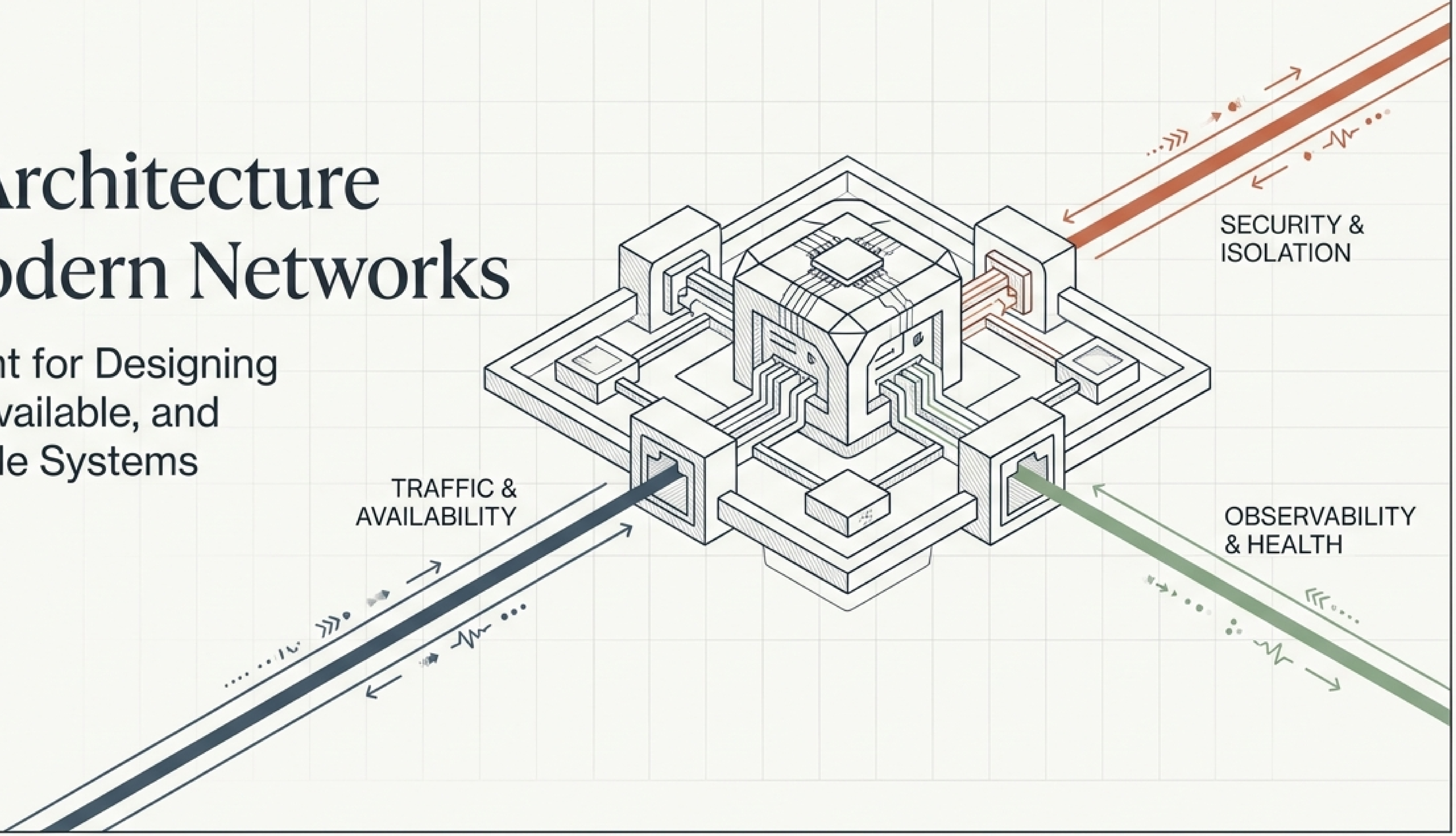
The Architecture of Modern Networks

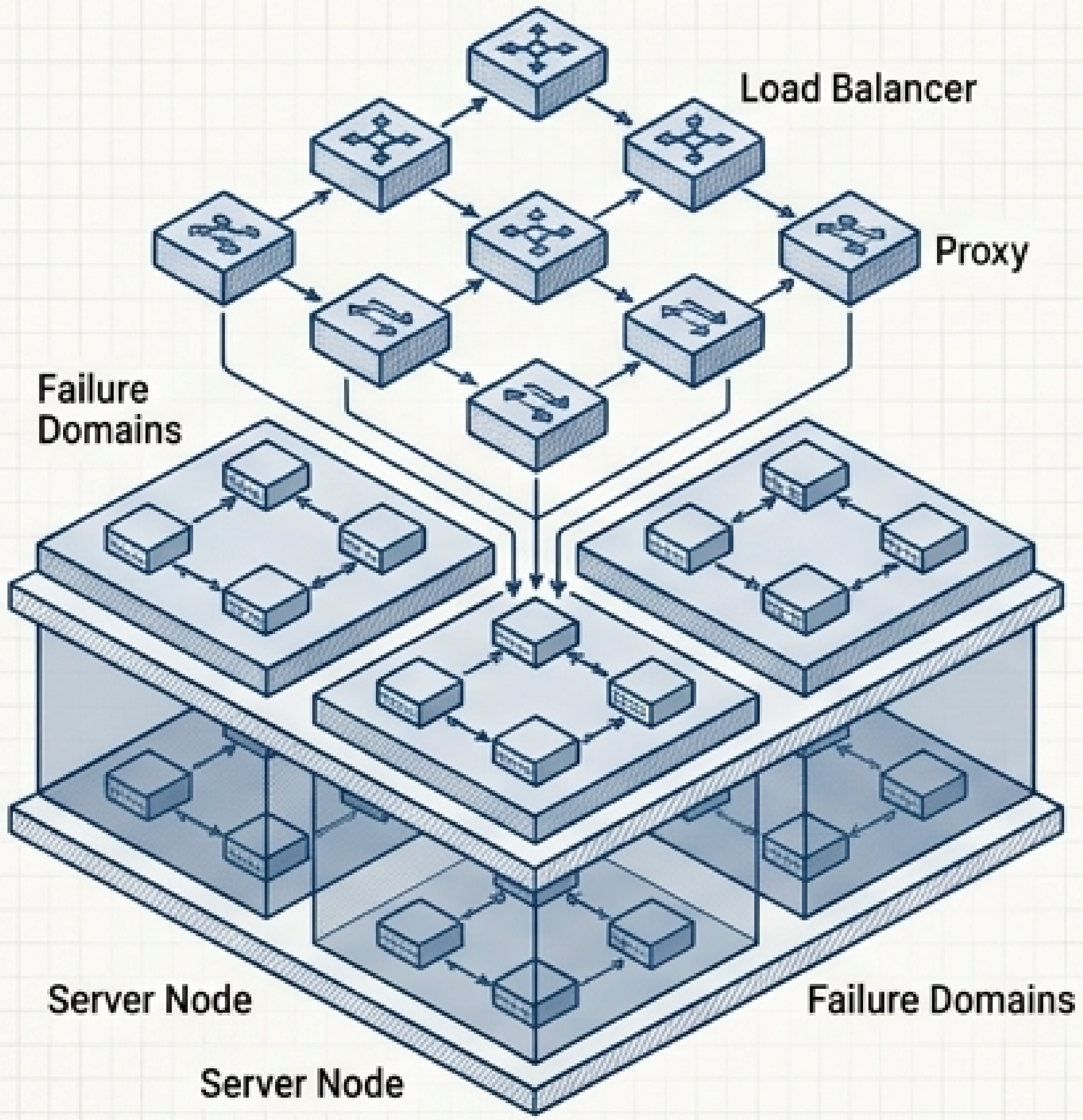
A Blueprint for Designing Secure, Available, and Observable Systems



TRAFFIC & AVAILABILITY

Focuses on proxies, load balancers, and highly available failure domains.

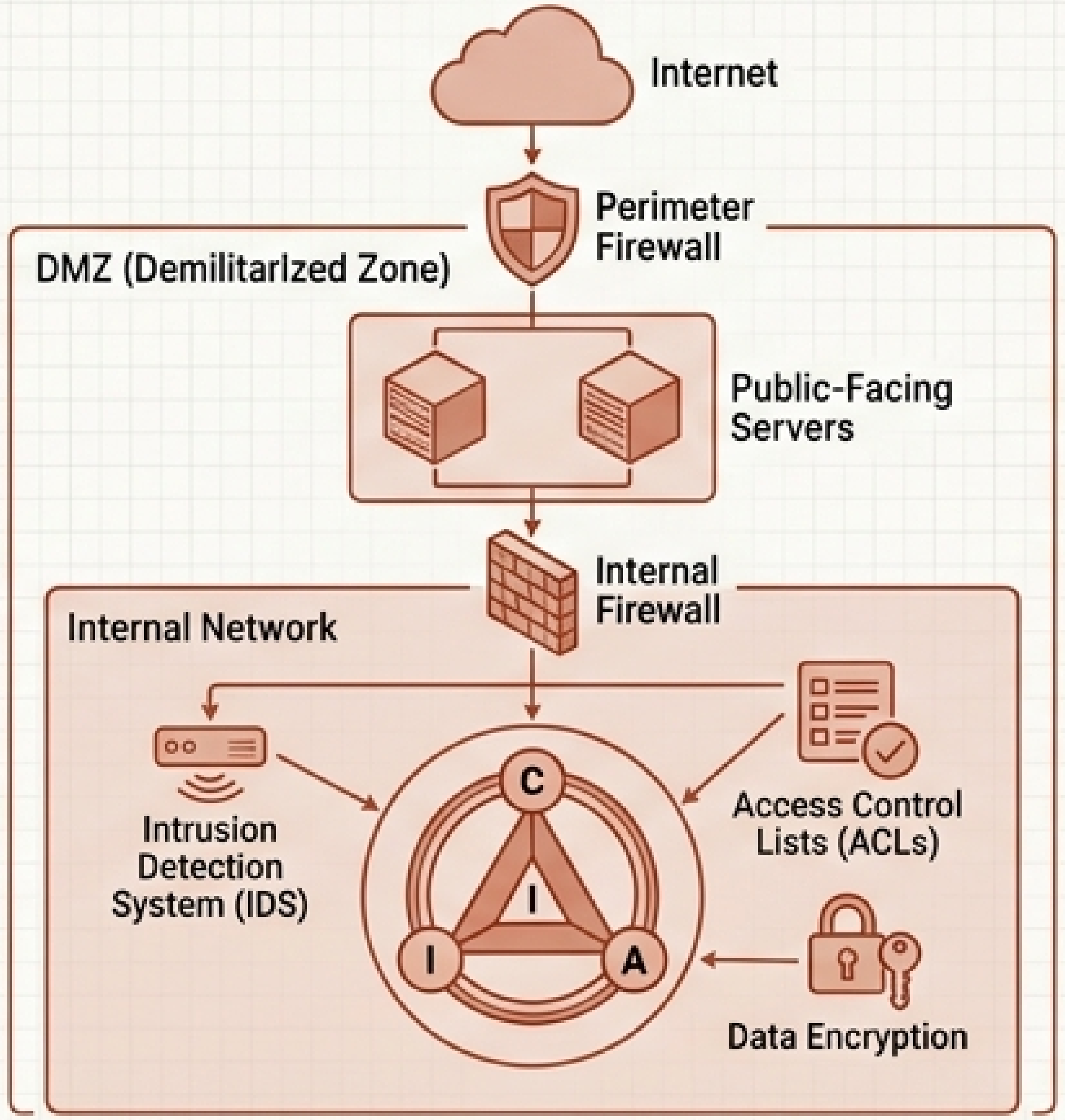
CORE GOAL: Ensure reliable, predictable communication.



SECURITY & ISOLATION

Focuses on firewalls, the DMZ, and defense in depth.

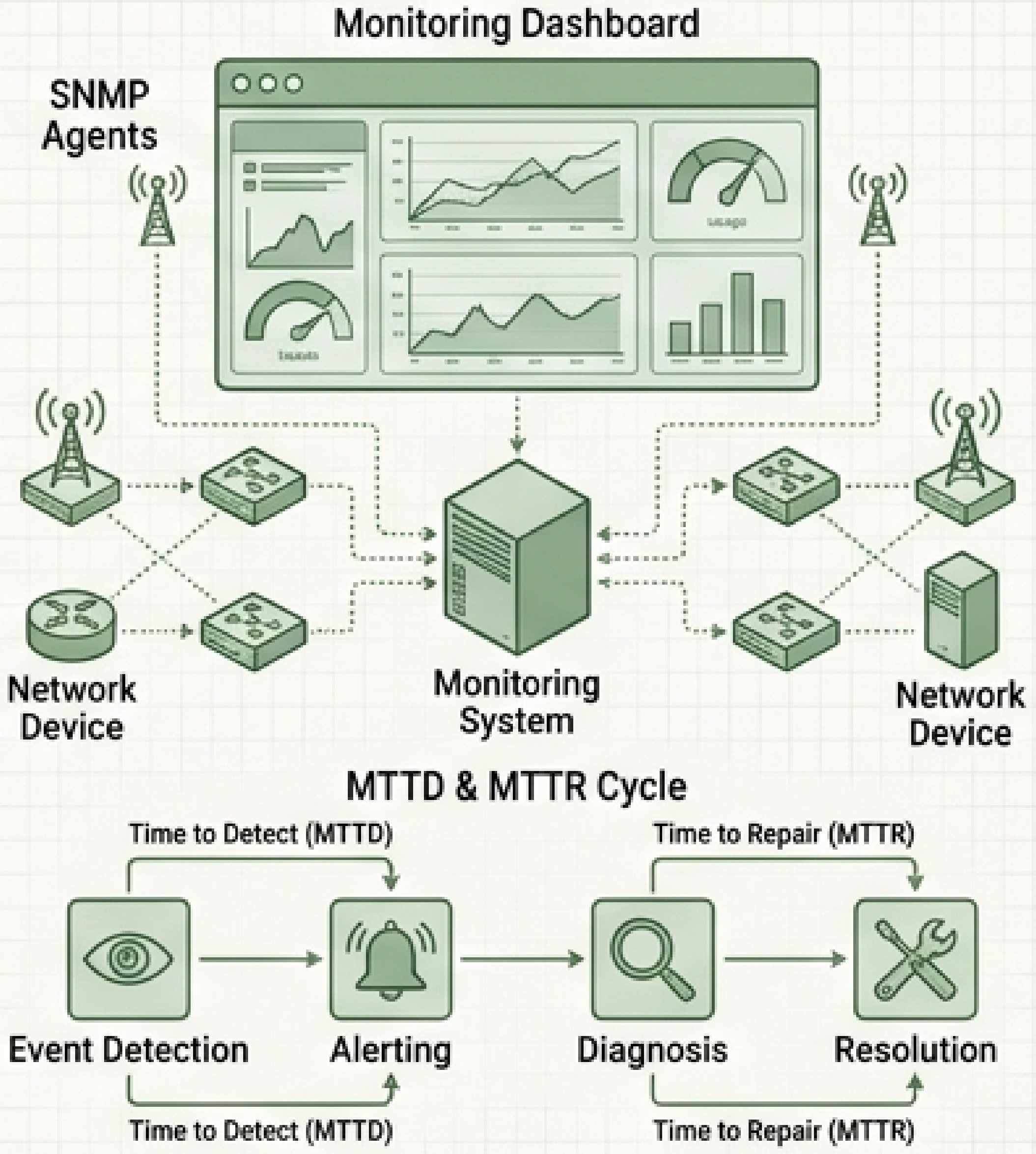
CORE GOAL: Protect confidentiality, integrity, and availability (CIA triad).



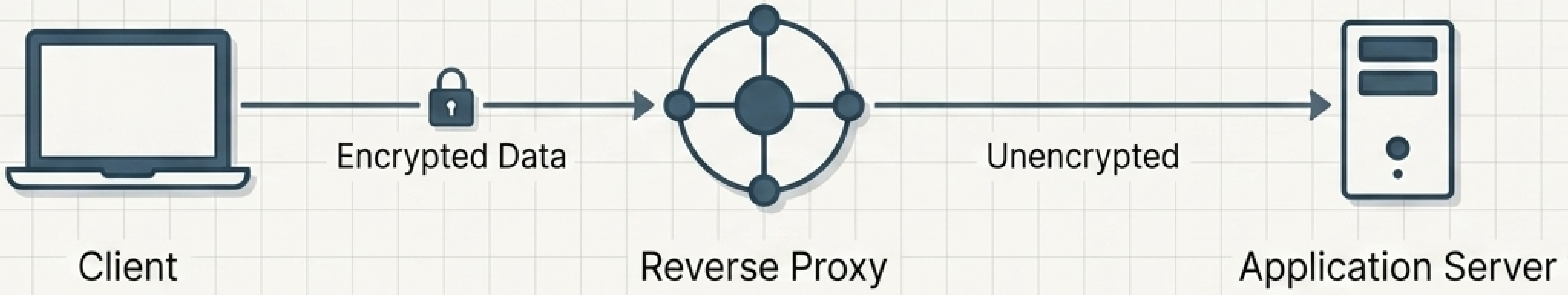
OBSERVABILITY

Focuses on monitoring, SNMP, and capacity planning.

CORE GOAL: Minimize Mean Time to Detect (MTTD) and Repair (MTTR).

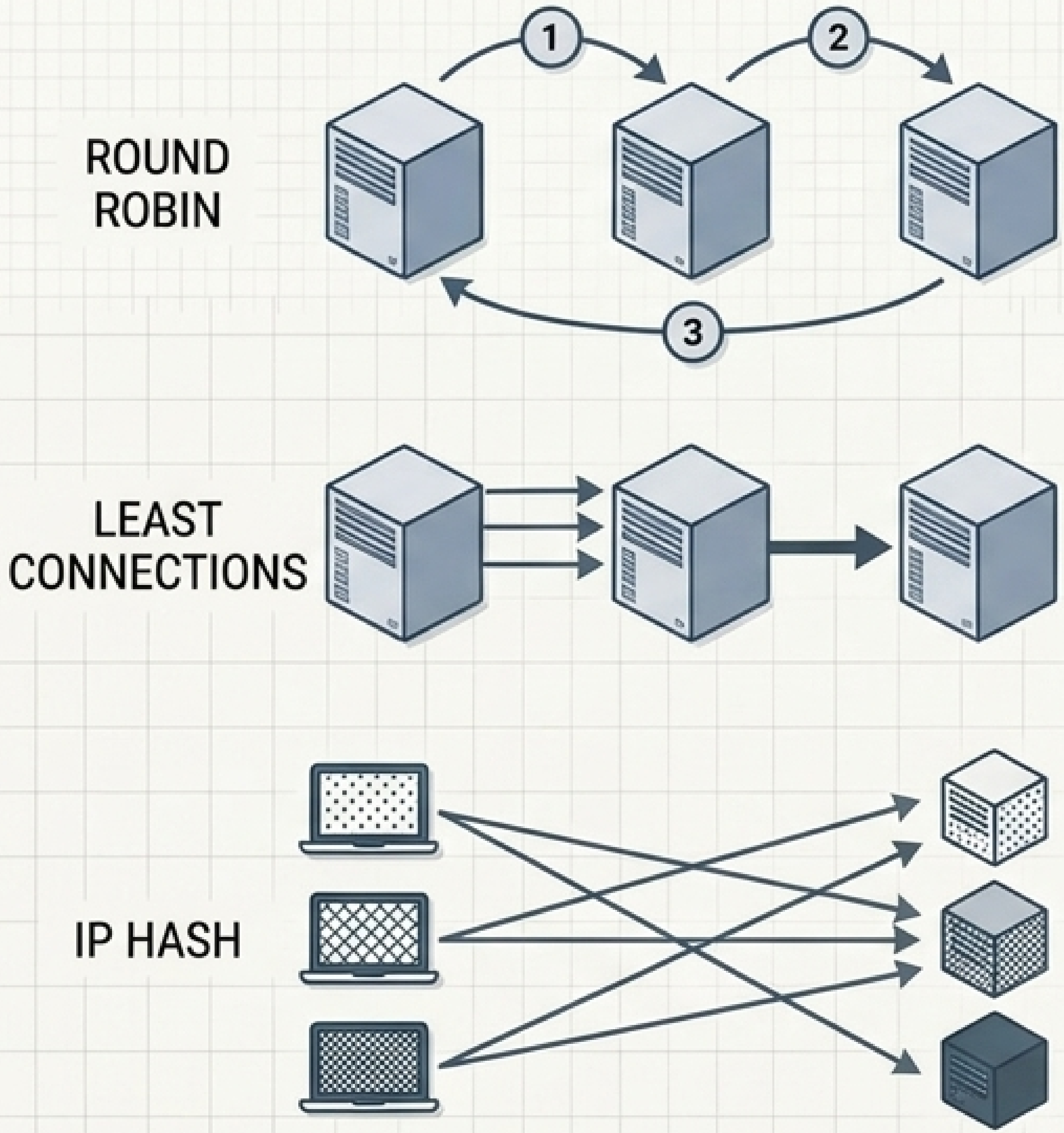


A professional network administrator does not treat these as isolated technologies. They form one living system.

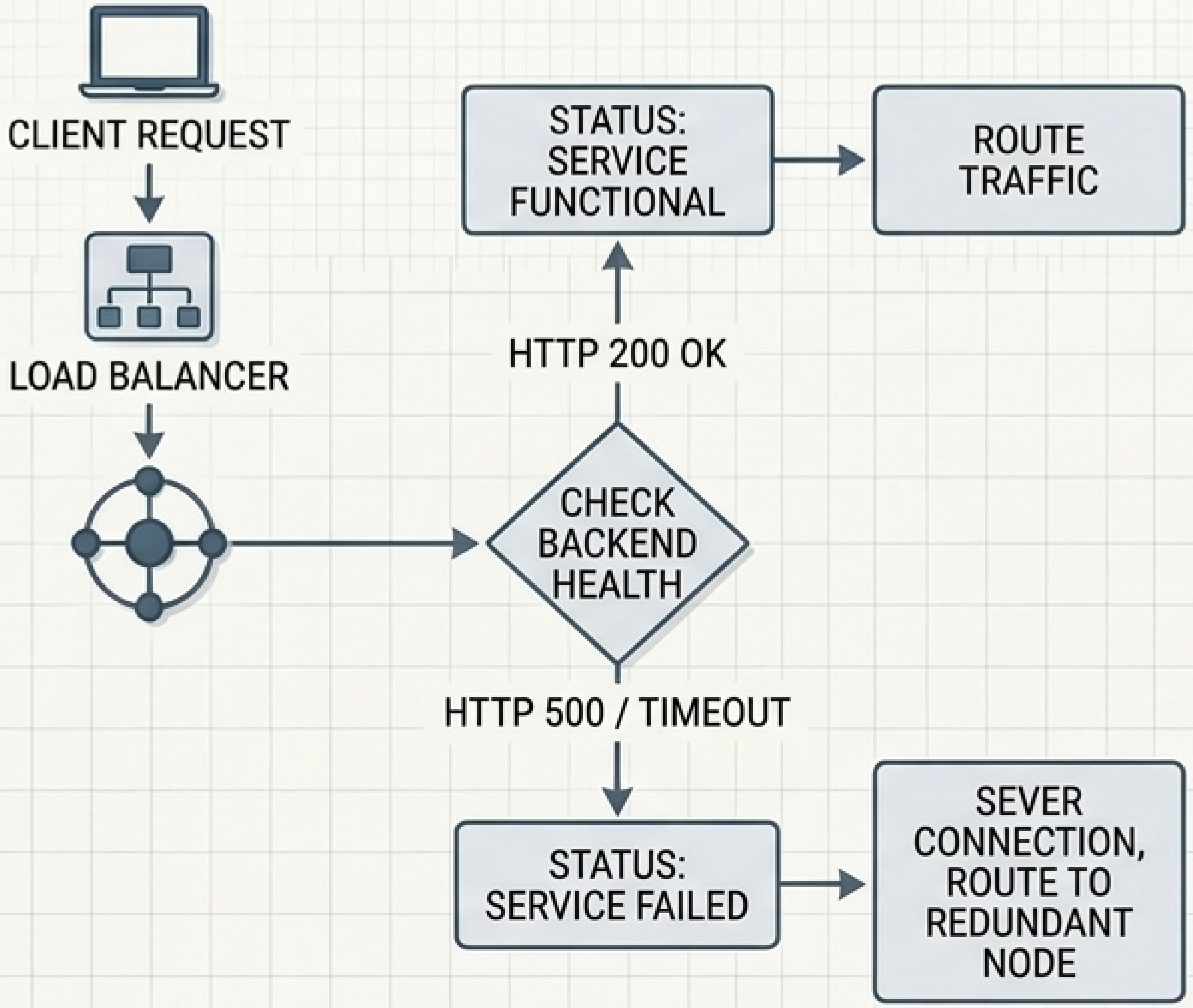


Forward Proxy	Reverse Proxy	Load Balancer
• Represents: The Client • Traffic: Client → Internet • Goal: Outbound access control, filtering, logging	• Represents: The Server/Application • Traffic: Internet → Application • Goal: Backend abstraction, centralized TLS termination, inbound security	• Represents: The Server Pool • Traffic: Internet → Multiple Backends • Goal: Scalability, fault isolation, resource utilization

ROUTING ALGORITHMS



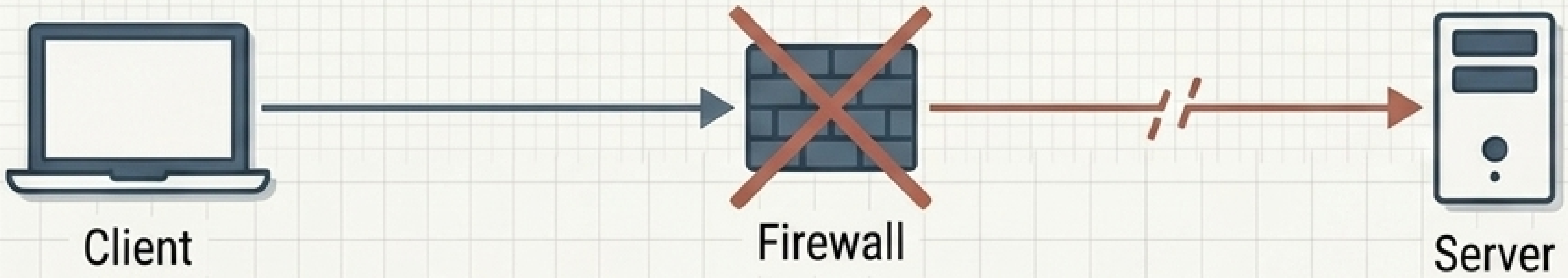
HEALTH CHECK LOGIC



Liveness (Ping) only proves the network is up. Application Health proves the service is functional.

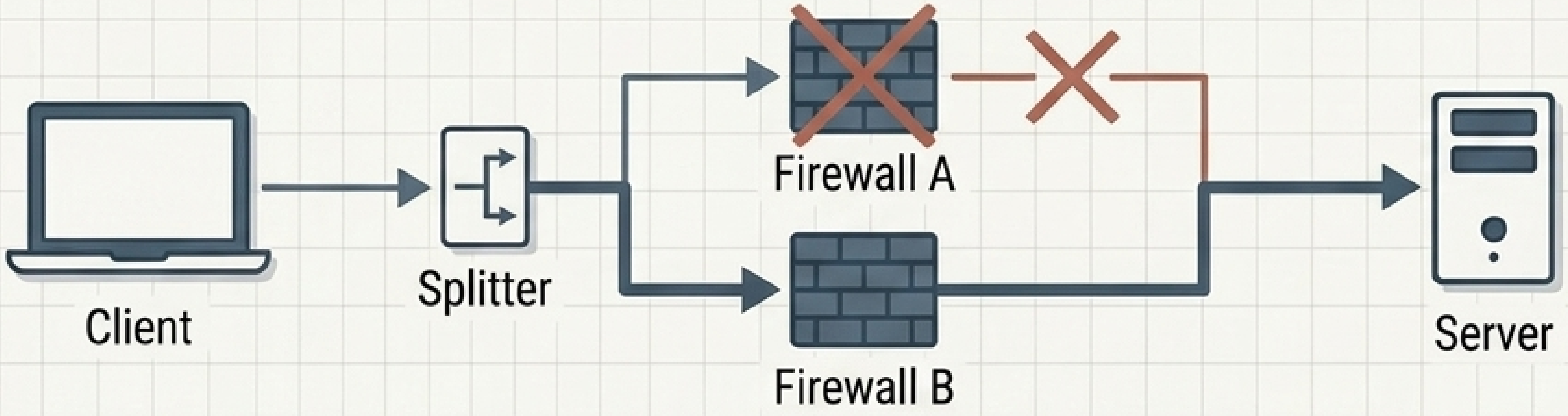
NETWORK AVAILABILITY: SPOF vs. HA ARCHITECTURE

THE SINGLE POINT OF FAILURE (SPOF)



Connection failed: No redundancy for the failed component.

THE HA ARCHITECTURE



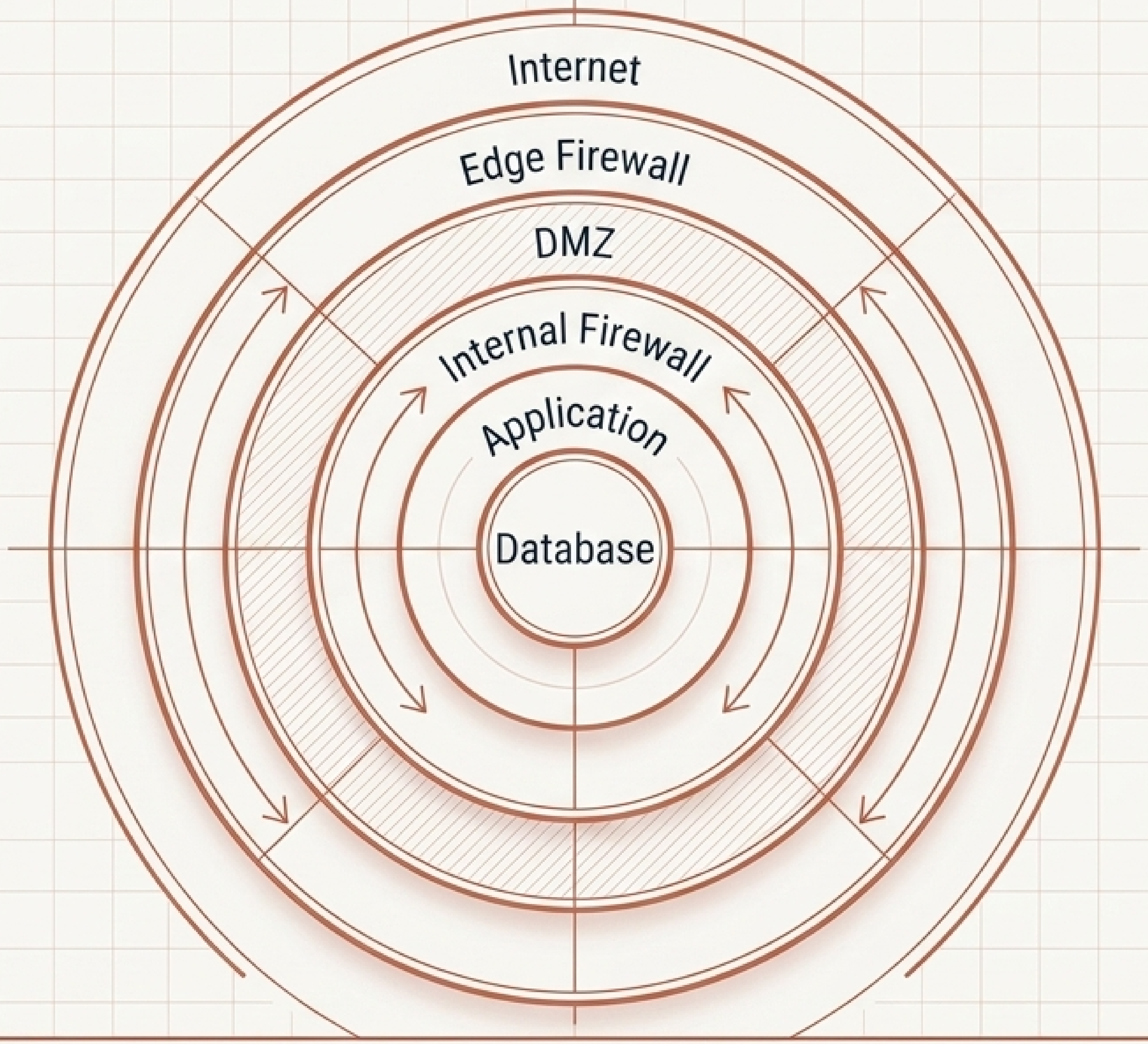
Connection successful: Traffic rerouted through redundant path.

ANNOTATIONS

High Availability (HA) requires failure detection, failover mechanisms, and redundancy.

The Failure Domain Trap: Redundancy fails if duplicate components share a single power source, building, or network path.

Best Practice: Ensure redundant components are geographically or physically isolated.

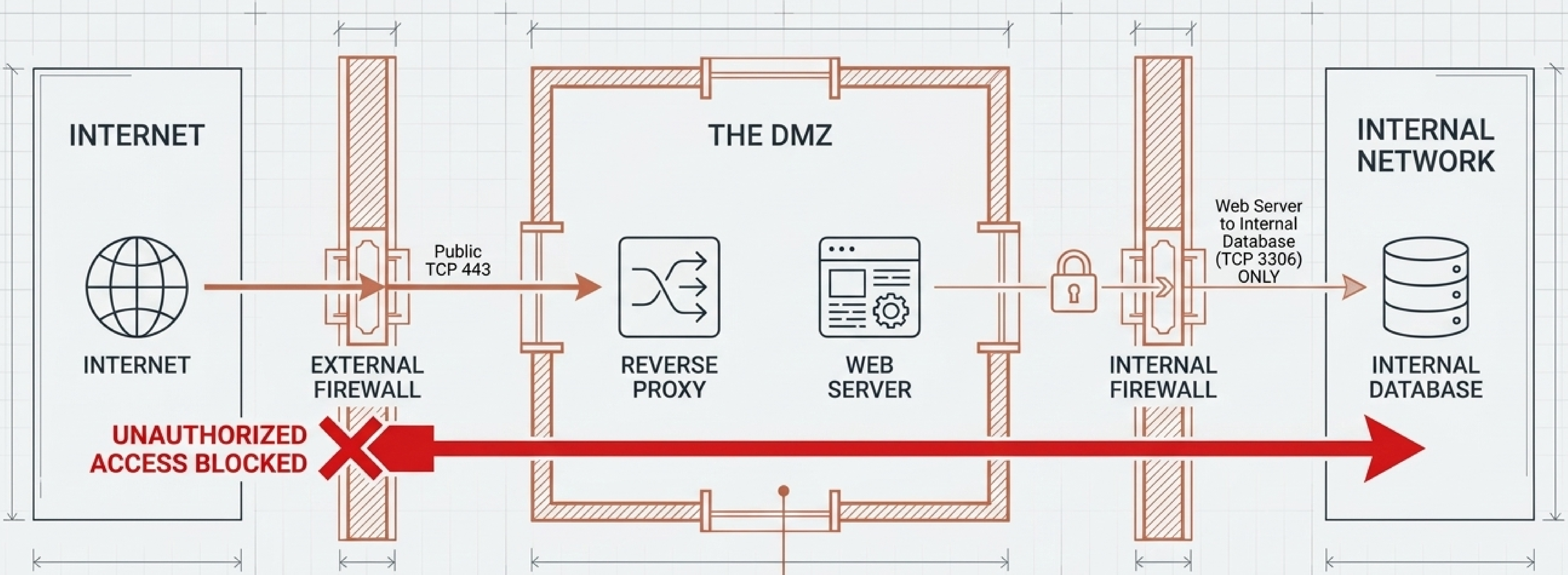


SECURITY OBJECTIVES (CIA TRIAD)

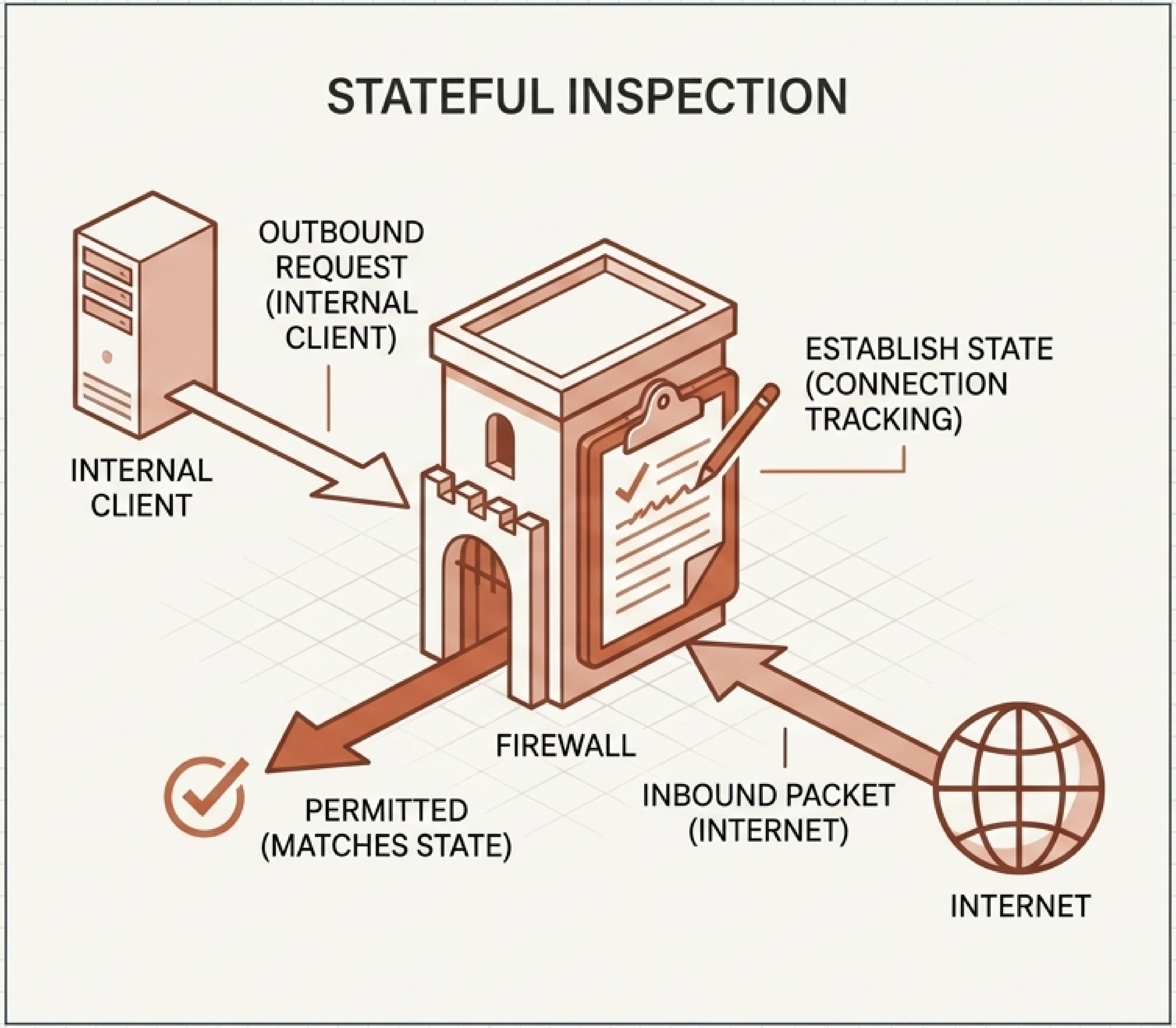
- **Confidentiality:** Prevent unauthorized disclosure.
- **Integrity:** Prevent unauthorized modification.
- **Availability:** Ensure authorized access.

Defense in Depth: Security must not rely on a single mechanism. If an outer perimeter fails, internal segmentation must remain intact to prevent full compromise.

DMZ: The Architectural Airlock



THE DMZ RULE: Never grant systems inside the DMZ unrestricted access to the internal network. Allow required communication; deny unnecessary communication.

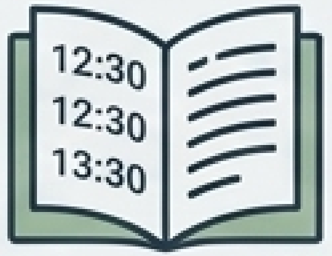


FIREWALL BEST PRACTICES

- Minimize exposure (open only required ports).
- Default Deny (explicitly allow, block everything else).
- Log important events for MTTR.
- Back up configurations before careful testing.

RULE ORDERING MATTERS: A broad 'Deny All' placed above an 'Allow HTTPS' rule will break the entire service.

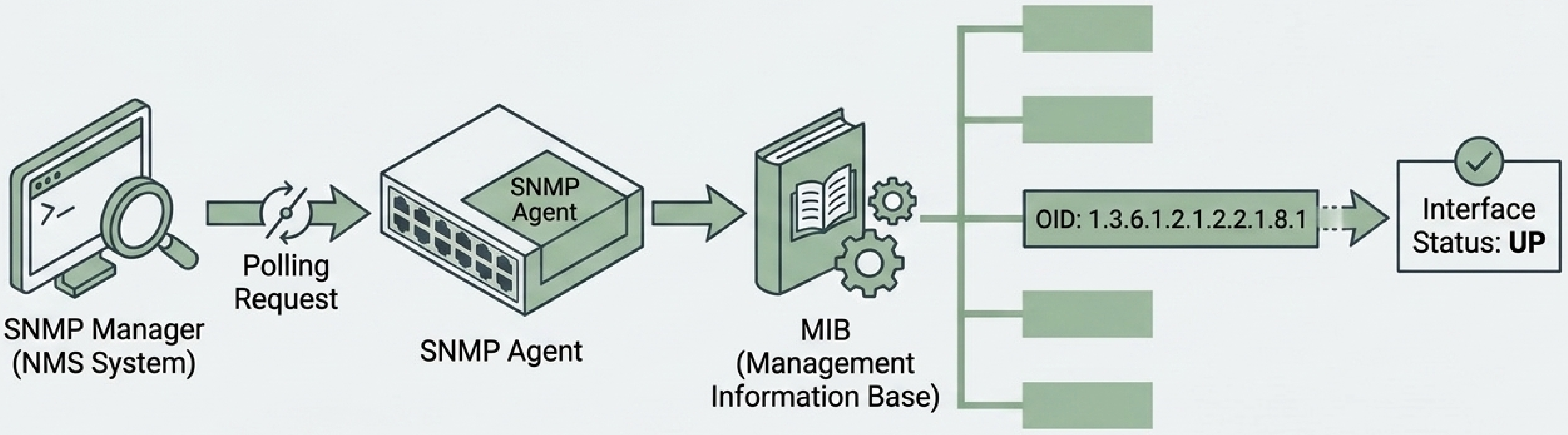
The Observability Matrix

1	 **Monitoring**	Answers "Is it healthy?" Focuses on Mean Time to Detect (MTTD). Uses ICMP, SNMP, and APIs to track real-time state.
2	 **Logging**	Answers "What happened?" Focuses on Mean Time to Repair (MTTR). Provides detailed, historical event context.

Alert Fatigue

Golden Rule: The goal is not to generate as many alerts as possible. The goal is to generate actionable alerts.

SNMP Architecture



Key Insight: A MIB is not a database file containing current values; it is a structural dictionary defining the manageable objects an Agent can expose to a Manager.

SNMP Communications Matrix

Mechanism	Direction	Purpose
Polling (GET / GETNEXT / GETBULK)	Manager → Agent	Periodically retrieve information to build historical data (e.g., CPU utilization every 5 mins).
TRAP	Agent → Manager	Unsolicited, immediate notification of a critical event (e.g., link failure, hardware fault).
INFORM	Agent ↔ Manager	An immediate notification that requires a confirmed acknowledgment from the manager

SNMP Security Tiers

Architectural Constraint: Restrict firewall ports UDP/161 and UDP/162 tightly.

SNMPv1 & v2c (Legacy)

Relies on Community Strings (functions like a shared password).

⚠ **Warning**: Often sent in cleartext. Do not treat community strings as modern cryptographic authentication. Restrict source IPs tightly.

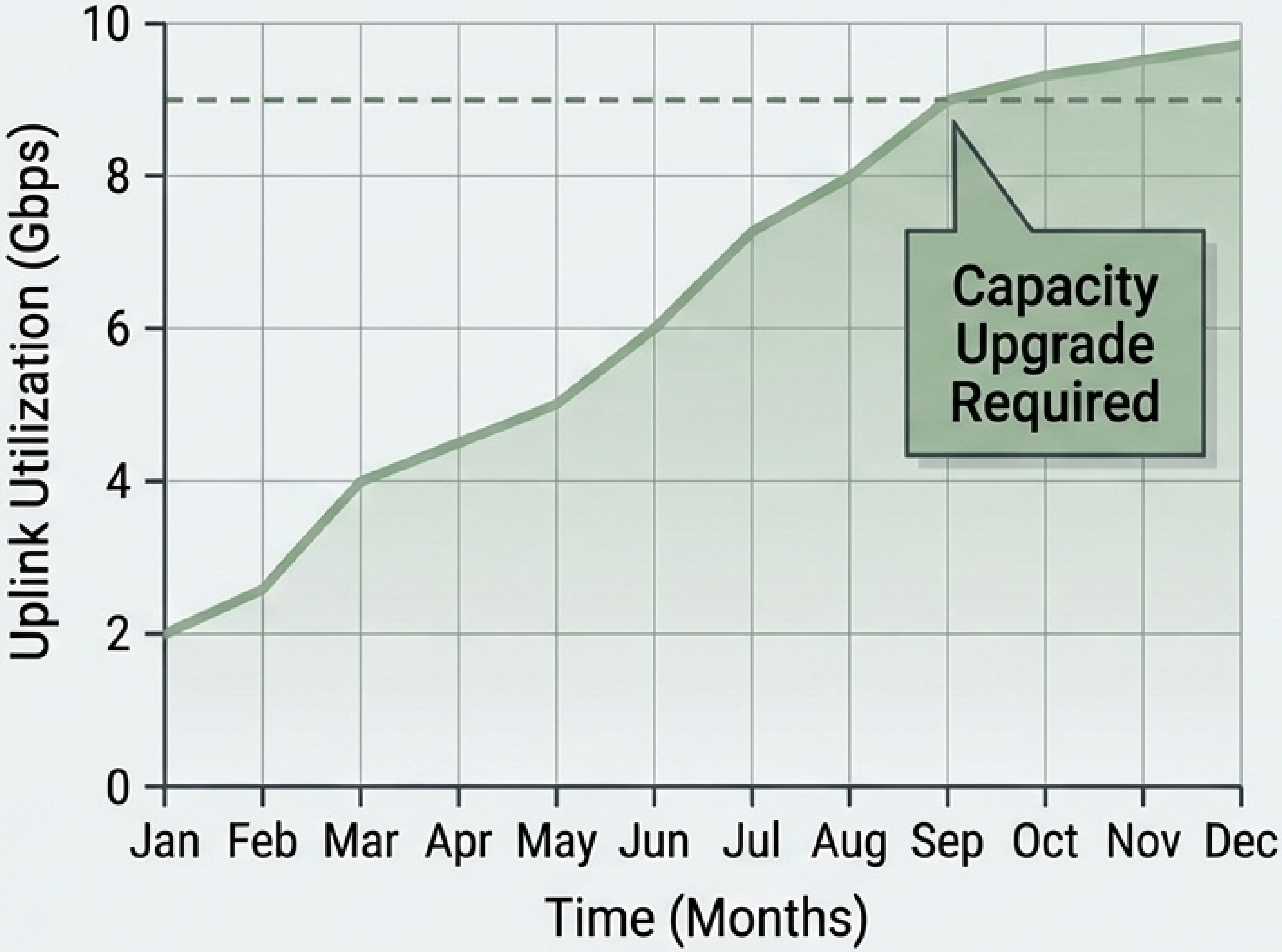
SNMPv3 (Modern/Preferred)

Adds three distinct security tiers:

1. **Authentication**: Verifies the sender.
2. **Integrity**: Ensures data wasn't altered in transit.
3. **Privacy**: Encrypts the payload.

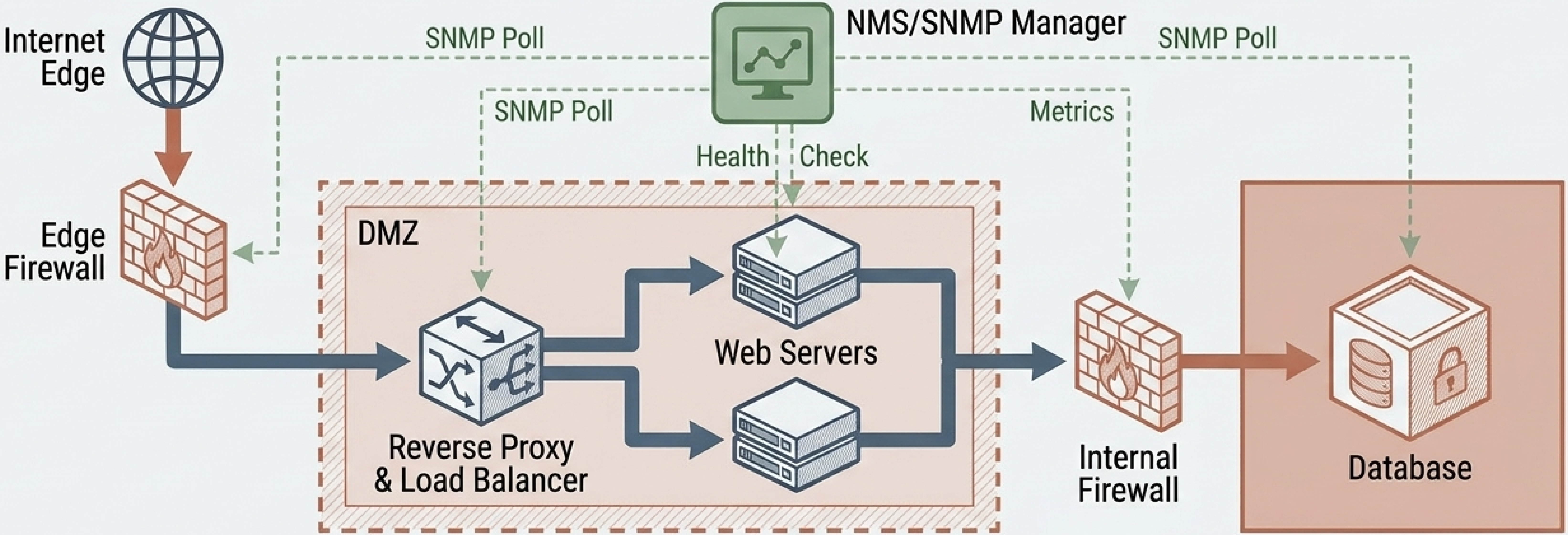
Rule: Always prefer v3 in security-sensitive environments.

Capacity Planning



- Monitoring is not just about detecting immediate failures; it is about plotting historical trends.
- By polling interface counters and discards over time, raw data turns into operational decisions.
- **Result**: Upgrading a congested link before packets drop preserves availability.

The Integrated Architecture Blueprint



An enterprise network is a single, integrated system designed for normal operation, failure tolerance, attack resistance, and rapid recovery.

Practical Troubleshooting Methodology

Troubleshoot from evidence, not assumptions. Don't immediately restart every server.

Layer 1 (Network): Is the host reachable? (Check IP, routing, ping).

Layer 2 (Security): Is the **required port reachable?** (Check firewalls, default deny policies).

Layer 3 (Intermediaries): Is the **Proxy/Load Balancer healthy?** (Check frontend listeners, health check states, TLS certificates).

Layer 4 (Backend): Is the **backend service healthy?** (Check resource usage, application logs, database dependencies).

Layer 5 (Application): Is the **application functional?** (Check HTTP status codes, specific query errors).